Q.1 .Differentiate Finite Automata, Push Down Automata and Turing Machine

Q.2. Discuss different applications of Finite Automata

Q.3. Explain with example Chomsky Hierarchy.

Q.4. Post Correspondence Problem.

Q.5. Recursive and Recursive enumerable languages.

Q.6. TM-Halting Problem.

Q.7. Difference b/w NFA and DFA

Q.8. Difference b/w Melay and Moore

Q.9. Pumping Lemma (Proof with Numerical) (CFL and Regular Language)

Q.10. Properties of Regular Language

Q.11. Rice's Theorem

Q.12 . Difference b/w NPDA and PDA

Q.13.Application of FA,PDA,TM

Q.14.

Decision Properties of Regular Languages

Post Correspondence Problem

Variants of Turing Machine

Acceptance by a PDA

Conversion of Moore to Mealy Machines

Q.15. Universal TM

Q.16. Differentiate Finite Automata, Push Down Automata and Turing Machine

Numericals

Q.1. Simplify the given grammar

S -> ASB | ϵ

A -> aAS | a

B -> SbS | A | bb

Q.2. Design DFA that accepts Strings with at least 3 a's. over Σ={a,b}.

Q.3. Design Moore machine for Σ={0,1}, print the residue modulo 3 for binary numbers.

Q.4. Design Push Down Machine that accepts L = {a^m b^n c^n d^m | m, n>0 }

Q.5. The grammar G is S ->aB | bA

A ->a| aS | bAA

B ->b | bS | aBB

Obtain parse tree for the following string “aababb” and check if the grammar is

Ambiguous.

Q.6. Explain Pumping Lemma with the help of a diagram to prove that given language is not a

regular language. L={0^m 1^m+1 | m>0}

Q.7. Design DFA that accepts Strings that ends in either “110” or “101” over Σ={0,1}.

Q.8. Design NFA that accepts strings starting with “abb” or “bba”

Q.9. Find Equivalent Greibach Normal Form (GNF) for given CFG.

S ->AA | a

A ->SS | b

Q.10. Define and design Turing Machine to accept 0n1n2n over Σ={0,1,2}.

Q.11. TM to Add Two Unary Number

Practice More Example And try to solve KT in this Attempt